WORKED SOLUTION

STEP I 1995, Question 14

Probability and Statistics • Estimating a Proportion

STEP I 1995, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s1-q14

The key idea

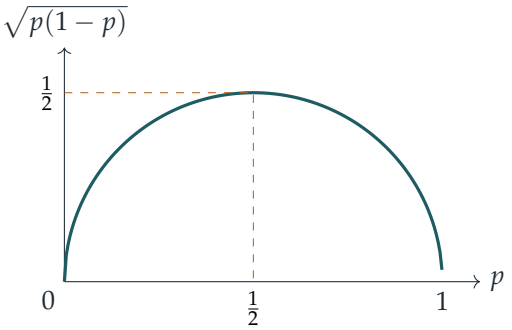
The sample proportion is a binomial count divided by n. For large n it is approximately normal with mean p and variance $p(1-p)/n$. The trouble is that the variance depends on the very p we don't know. Part (i) solves that: it gives the worst possible value of the standard deviation, so a sample size chosen for the worst case works whatever p is. Part (iii) then asks what changes when we already have a rough idea of p.

Solution

Part (i)

Let $f(p) = p(1-p) = \frac{1}{4} - \left(p - \frac{1}{2}\right)^2$, by completing the square. So $f(p) \leq \frac{1}{4}$ on $[0,1]$, with equality only at $p = \frac{1}{2}$. The square root is increasing, so

$$\max_{0 \leq p \leq 1} \sqrt{p(1-p)} = \sqrt{\tfrac{1}{4}} = \tfrac{1}{2}, \qquad \text{attained at } p = \tfrac{1}{2}.$$



Part (ii)

Let X be the number of females in the sample. Each person picked is female with probability p. We assume the population is large, so that the n choices are effectively independent (sampling without replacement from a huge population is almost the same as sampling with replacement). Then

$$X \sim \mathrm{B}(n,p), \qquad \hat{p} = \frac{X}{n}, \qquad \mathrm{E}(\hat{p}) = p, \qquad \mathrm{Var}(\hat{p}) = \frac{np(1-p)}{n^2} = \frac{p(1-p)}{n}.$$

For large n the central limit theorem (the normal approximation to the binomial) gives

$$\hat{p} \approx \mathrm{N}\left(p,\ \frac{p(1-p)}{n}\right), \qquad \text{so} \qquad Z = \frac{\hat{p}-p}{\sqrt{p(1-p)/n}} \approx \mathrm{N}(0,1).$$

We want $\mathrm{P}(|\hat{p}-p| > 0.01) < 0.01$, that is,

$$\mathrm{P}\left(|Z| > \frac{0.01\sqrt{n}}{\sqrt{p(1-p)}}\right) < 0.01.$$

By symmetry this is 1% split as 0.5% in each tail. Since $\Phi(2.576) = 0.995$, we need

$$\frac{0.01\sqrt{n}}{\sqrt{p(1-p)}} > 2.576 \quad \Longleftrightarrow \quad \sqrt{n} > 257.6\sqrt{p(1-p)}.$$

Ask yourself We don't know p, so how can we choose n from an inequality that involves it?

This is where part (i) comes in. Whatever p is, $\sqrt{p(1-p)} \le \frac{1}{2}$. So if $\sqrt{n} > 257.6 \times \frac{1}{2} = 128.8$, the inequality holds for every p. This condition is also necessary, because when $p = \frac{1}{2}$ (and the proportion of females is close to that) the bound is attained. Therefore we need

$$n > 128.8^2 = 16\,589.44,$$

and the required sample size is

$$\boxed{n = 16\,590} \qquad (\text{about } 16\,600).$$

With a more precise percentage point, $z = 2.5758$, we get $n > 16\,587.2$. The answer is "about 16 600" either way, and the third significant figure depends on the tables used.

Part (iii)

In general the requirement is

$$n > \left(\frac{2.576}{0.01}\right)^2 p(1-p) \approx 66\,360\,p(1-p).$$

The value 16 590 came from the worst case $p = \frac{1}{2}$. If we know that p lies well away from $\frac{1}{2}$, we can use a smaller bound for $p(1-p)$.

(a) Left-handedness. Roughly 10% of people are left-handed. If we may safely assume $p \leq 0.15$, then $p(1-p) \leq 0.15 \times 0.85 = 0.1275$, since $p(1-p)$ is increasing on $[0, \frac{1}{2}]$. This gives $n > 8461$. If $p = 0.1$ exactly, $n > 66\,360 \times 0.09 \approx 5970$, so about 6000. Either way the sample is much smaller than the 16 590 needed when we know nothing. The normal approximation is still good, since np is in the hundreds.

(b) Millionaires. Now p is very small, perhaps 0.001 to 0.01. Taken literally, $p \leq 0.01$ gives $p(1-p) \leq 0.0099$ and $n > 657$. But this misses the point:

- An error of 0.01 is useless when p itself is about 0.001: the estimate $\hat{p} = 0$ already lies within 0.01 of the truth. What we really want is a *relative* accuracy, say an error of at most 1% of p. Then we need $n > 2.576^2(1-p)/(p \times 0.01^2)$, which for $p = 0.001$ is about 6.6×10^7. That is enormously larger.
- With p tiny and n only a few hundred, np is less than 1. The binomial is then very skewed, and the normal approximation is no longer valid; a Poisson approximation would be the better model.
- In practice millionaires are hard to reach, and people may not tell the truth about their wealth, so random sampling and honest responses are both doubtful.

So for left-handedness, knowing p roughly lets us use a smaller sample. For millionaires, any sensible accuracy requirement needs a far larger sample, or a different sampling method altogether.

Checks

- At $p = \frac{1}{2}$ and $n = 16\,590$ the normal approximation gives $\mathrm{P}(|\hat{p} - p| > 0.01) = 2\big(1 - \Phi(2.5762)\big) \approx 0.00999$, just under 1%.
- Summing the exact binomial probabilities for $n = 16\,590$, $p = \frac{1}{2}$ gives about 0.0102. The small excess comes from the discreteness of the binomial (the normal approximation without continuity correction). With $p = 0.3$ the exact value is about 0.005, comfortably inside, as part (i) predicts.
- For $p = 0.1$ and $n = 6000$ the exact binomial probability is about 0.0098, which confirms the figure in (iii)(a).

Common slips

- Using 1.96 or 2.326 instead of 2.576. The 1% is split between the two tails, so we need $\Phi(z) = 0.995$.
- Leaving n in terms of p. The question wants a single number that works for any p, which is why part (i) is there.
- Reading "$\sqrt{p(1-p)} \leq \frac{1}{2}$" as giving $n > 128.8$. It is $\sqrt{n}$, not n, that must exceed 128.8.

Beyond STEP

This is the standard calculation behind the "margin of error" quoted in opinion polls. A poll of about 1000 people gives roughly ± 3 percentage points at 95% confidence, by exactly the worst-case argument of part (i). At university it leads to confidence intervals, and to the question of how to estimate rare proportions, where Poisson and exact binomial methods take over from the normal approximation. Blitzstein and Hwang, *Introduction to Probability*, covers the central limit theorem and these approximations very readably.

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